Breno Livio Silva de Almeida

Computational biology · Biological sequence representation · Machine learning

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Website Google Scholar 0000-0002-2041-9170 GitHub LinkedIn

Research interests: Representation and prediction of biological sequences, combining handcrafted descriptors, learned representations from foundation models, metagenomic metadata curation and automated machine learning for classification and annotation in microbial ecology.

Education

2024 – Present **PhD in Computer Science — University of São Paulo / Leipzig University.** Double-degree program. Thesis: *Decoding the Language of Life: Towards the Limits of Biological Sequence Representation*. Supervisors: André C. P. L. F. de Carvalho and Peter F. Stadler.

2020 – 2024 **BSc in Computer Science — University of São Paulo.** GPA: 9.4/10; rank: 7/116. Specializations: Data Engineering and Data Science. Thesis: *BioDeepFuse: A Hybrid Deep Learning Approach with Handcrafted Features for Non-coding RNA Classification*. Research conducted with UFZ, supported by the São Paulo Research Foundation (FAPESP); supervisor: André C. P. L. F. de Carvalho.

Research Experience

07.2025 – 06.2026 **Research Assistant — Helmholtz Centre for Environmental Research (UFZ).** Microbial Data Science group, Department of Computational Biology and Chemistry.

- Contributed to protein-language-model approaches for functional annotation of metagenomic sequences within AI MareExplore.
- Contributed to AI-assisted data-type categorization and ontology alignment in EReMiD.

07.2022 – 09.2022 **Research Intern — Big Data Biology Lab, Fudan University.** Performed rarefaction analyses of small open reading frames across environmental metagenomes, contributing to *A catalog of small proteins from the global microbiome* (*Nature Communications*, 2024).

International Collaborations

Dates indicate my participation; budgets refer to projects or institutions, not individual awards.

2024 – Present **AutoAI-Pandemics.** AI4PEP collaboration across Brazil, Germany, Canada, and the UK developing accessible AI for epidemic preparedness and biological-data analysis. **CAD 362,500 awarded in 2023** for four years, supported by IDRC and UK International Development. I contribute to BioAutoML-FAST and publications on AutoAI Hub and Dominique. AI4PEP Brazil hub · Funding announcement.

2024 – Present **NFDI4Microbiota.** German consortium providing FAIR microbial data, analysis tools, and training. Initial DFG funding: **approximately EUR 3 million annually for five years from October 2021**, including **approximately EUR 1 million for UFZ** over that period. I contribute to microbial genome and metadata deposition and workshop teaching. Funding and UFZ role · DFG project record.

2025 – Present **AI MareExplore.** Helmholtz Innovation Pool project (2025–2027), led by GEOMAR with UFZ, Forschungszentrum Jülich, and GFZ, using AI to discover marine microbial enzymes for plastic degradation and carbon capture. I contribute protein-language-model approaches for microbial protein annotation. Funder, dates, and partners · Project announcement.

Research Funding and Fellowships

Awarded approximately **€110,181** in competitive fellowships, research scholarships, and mobility/travel grants across the award periods.

2025 **Google PhD Fellowship in Health Research** — USD 30,000. International fellowship recognizing outstanding doctoral research in computing and related fields, with USD 15,000 annually for two years and mentorship from a Google researcher. Selected among 15 fellows in Latin America in 2025, including eight from Brazil; supports research on language models for biological sequences. Recipients · USP award report.

2025 **ISME Scholar Mobility Fund** — EUR 2,300. Awarded by the International Society for Microbial Ecology for a research visit to UFZ, Leipzig, in July 2026. The program supports international research and training visits by graduate students and postdoctoral researchers to advance microbial ecology and collaboration with host laboratories. Program · USP visit report.

2024 **FAPESP Doctoral Scholarship** — BRL 453,168. Competitive, peer-reviewed scholarship from the São Paulo Research Foundation (FAPESP), grant #2024/10958-1. Supports the project *Decoding the language of life: Towards protein classification and design through large language models*, investigating alignment-free sequence representations and machine learning for protein classification and design at ICMC-USP. FAPESP grant record.

2024 **FAPESP Technical Training Scholarship** — BRL 10,553.20. Grant #2024/03304-5, April–October 2024. Funded *Democratization of machine learning in life sciences and annotation of protein families*, in collaboration with UFZ. The project develops accessible web applications for RNA classification and AutoML, and explores language models for protein-family annotation; linked to the CeMEAI research center at USP. FAPESP grant record.

2024 **Prototypes for Humanity Travel Grant** — USD 760. Travel support to present BioAutoML at Prototypes for Humanity in Dubai, November 2024. The project was selected for an international showcase of university innovations addressing social and environmental challenges, providing an opportunity to present accessible machine-learning tools for life-science research. Selected project.

2023 **FAPESP Scientific Initiation Scholarship** — BRL 7,920.99. Competitive, peer-reviewed scholarship; grant #2023/00264-0, April–December 2023. Funded undergraduate research on *Machine learning applied to multiomics analysis for exploration of microbial communities*, examining how methods including BioAutoML can integrate omics data into analysis workflows for complex microbial communities. FAPESP grant record.

2023 **CNPq Scientific Initiation Scholarship** — BRL 1,500. National Council for Scientific and Technological Development, grant #101243/2023-5. Supported undergraduate research on *Automated machine learning for life sciences*; scientific-initiation scholarships develop research skills through supervised participation in a scientific project. CNPq scholarship category.

2021 – 2022 **FAPESP Scientific Initiation Scholarship** — BRL 7,652.70. Competitive, peer-reviewed scholarship; grant #2021/08561-8, September 2021–June 2022. Funded *Classification and analysis of bacterial non-coding RNA sequences using machine-learning techniques*, combining mathematical sequence descriptors and predictive models to distinguish small non-coding RNAs and evaluate informative features. FAPESP grant record.

Indicative total at fixed reporting rates: BRL 1 = EUR 0.17; USD 1 = EUR 0.85. These are indicative conversions, not current market rates. Total excludes cash prizes, estimated training values, and consortium budgets; it represents award values, not payments received.

Honors and Awards

2026 **IRCAI Global Top 100 — Outstanding (2025 edition).** Team recognition for BioAutoML from the International Research Centre on Artificial Intelligence (IRCAI), under the auspices of UNESCO. Recognizes accessible machine learning for life sciences supporting UN Sustainable Development Goals 3 (health) and 10 (reduced inequalities). Contributing developer of BioAutoML; currently lead developer of BioAutoML-FAST. Official recognition.

2026 **HIDA-sponsored Carpentries Instructor Training.** Selected for one of 15 annually sponsored places for Helmholtz researchers; estimated training value: USD 1,600. The program prepares researchers to teach practical computing and data-science workshops using evidence-based instruction, supporting reproducibility and open science. Participation leads toward Carpentries instructor certification. HIDA program and selection details.

2024 **J. F. Marar Artificial Intelligence Prize for Undergraduate Research.** Winner for *BioDeepFuse: Empowering Researchers in Life Sciences with Deep Learning*; cash prize: **BRL 10,500**. Inaugural award for ICMC-USP undergraduate AI research (2023 edition, presented in February 2024), selected unanimously by the committee on academic relevance, social relevance, alignment with the UN Sustainable Development Goals, and the student's academic performance. BioDeepFuse combines deep learning and handcrafted sequence features to make non-coding RNA classification more accessible to life-science researchers. USP award announcement.

2024 **Prototypes for Humanity.** BioAutoML selected among 100 projects from nearly 3,000 submissions worldwide. Presented in Dubai in November 2024, showcasing automated machine learning that enables life-science researchers to analyze biological sequences with less programming expertise. Travel support is reported in Research Funding and Fellowships. Selected project.

Publications

h-index: 4 **i10-index:** 3 **Citations:** 119 (Google Scholar), 74 (Scopus)

Google Scholar as of 2026-09-18 · Scopus as of 2026-09-18.

† Equal contribution.

Published Journal Articles

- Duan, Y., Santos-Júnior, C. D., Schmidt, T. S., Fullam, A., **de Almeida, B. L.**, Zhu, C., … & Coelho, L. P. (2024). A catalog of small proteins from the global microbiome. Nature Communications, 15(1), 7563. **DOI:** 10.1038/s41467-024-51894-6.
- **de Almeida, B. L.**†, Avila Santos, A. P.†, Bonidia, R. P., Stadler, P. F., Stefanic, P., Mandic-Mulec, I., … & de Carvalho, A. C. (2024). BioDeepfuse: a hybrid deep learning approach with integrated feature extraction techniques for enhanced non-coding RNA classification. RNA Biology, 21(1), 410-421. **DOI:** 10.1080/15476286.2024.2329451.
- Bonidia, R. P., Avila Santos, A. P., **de Almeida, B. L.**, Stadler, P. F., Nunes da Rocha, U., Sanches, D. S., & De Carvalho, A. C. (2022). Information theory for biological sequence classification: A novel feature extraction technique based on Tsallis entropy. Entropy, 24(10), 1398. **DOI:** 10.3390/e24101398.
- Bonidia, R. P., Santos, A. P. A., **de Almeida, B. L.**, Stadler, P. F., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. (2022). BioAutoML: automated feature engineering and metalearning to predict noncoding RNAs in bacteria. Briefings in Bioinformatics, 23(4), bbac218. **DOI:** 10.1093/bib/bbac218.

Accepted Journal Articles

- **de Almeida, B. L. S.**†, Bonidia, R.†, Florentino, B.†, Barreto, M. C. A.†, Constantino, J. L. R.†, Tavares da Silva, D.†, Zwicker, G. G., Rocha, U., Goedert, G., Struchiner, C., & de Carvalho, A. AutoAI Hub: From data to society, democratizing machine learning for everyone. Oxford Open Digital Health. **Accepted for publication**.
- Tavares, D., Barreto, M. C. A., **de Almeida, B. L. S.**, Florentino, B. R., Tetzner, F., Goedert, G., Struchiner, C., Bonidia, R., & de Carvalho, A. Dominique: An AI-powered fact-checking chatbot for democratizing access to reliable information. Oxford Open Digital Health. **Accepted for publication**.

Conference Proceedings

- **de Almeida, B. L. S.**†, Queiroz, A. P.†, Santos, A. P. A.†, Bonidia, R. P., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. P. D. L. F. (2021, November). Feature importance analysis of non-coding DNA/RNA sequences based on machine learning approaches. In Brazilian Symposium on Bioinformatics (pp. 81-92). Cham: Springer International Publishing. **DOI:** 10.1007/978-3-030-91814-9_8.

Preprint

- **de Almeida, B. L.**, Bonidia, R. P., Bole, M., Avila-Santos, A. P., Stadler, P. F., Nunes da Rocha, U., & de Carvalho, A. C. P. L. F. (2026). BioAutoML-FAST: an automated machine-learning platform for reusable and benchmarked biological sequence models. *bioRxiv*. **DOI:** 10.64898/2026.04.18.719383.

Research Software and Resources

BioAutoML-FAST	**Automated machine learning for biological sequences.** Lead developer; expanded sequence types, prediction tasks, and evaluation within the BioAutoML framework. Code · Web application · Preprint. Provides reusable trained models for genomics, transcriptomics, and proteomics.
BioAutoML	**Automated feature engineering and model selection.** Contributing developer. Automates sequence-feature extraction and selection, algorithm recommendation, and hyperparameter tuning for biological sequence classification, including bacterial non-coding RNAs. Code · Paper.
BioDeepFuse	**Hybrid deep learning for non-coding RNA classification.** Co-first author of the methods paper. Combines convolutional or bidirectional LSTM networks with handcrafted sequence features to capture complementary sequence patterns. Includes training and evaluation scripts, datasets, and experimental results for bacterial RNA classification. Code · Paper.
MAG2ENA	**Metagenome-assembled genome submission to the European Nucleotide Archive.** First co-author; developed within NFDI4Microbiota. Provides web and command-line workflows for metadata curation, validation, and genome submission using the MIMAG checklist, plus metadata updates for existing ENA records. Code · Web application.

Invited Talks and Conference Presentations

Invited Talk

- **de Almeida, B. L.** Da bancada ao teclado: ferramentas gratuitas para a pesquisa científica [From bench to keyboard: free tools for scientific research]. **Invited talk**, 36th FisFar Summer Course, Federal University of Minas Gerais (UFMG), Brazil. January 2027. *Scheduled.*

Oral Presentations and Flash Talks

- **de Almeida, B. L.**, Kamath, S., Santos, A. P. A., Bonidia, R. P., Espíndola-Hernández, P., Stadler, P. F., Wierckx, N., Borchert, E., Rother, D., Oliveira, A., de Carvalho, A. C. P. L. F., & Rocha, U. N. Combining protein language models with different architectures expands protein family annotation 14.2-fold across 16 million open reading frames from 3,802 environmental metagenomes. **Oral presentation**, 20th International Symposium on Microbial Ecology (ISME20), Auckland, New Zealand, 18 August 2026 (MS13, Computational Approaches in Microbial Ecology). Conference program.
- **de Almeida, B. L. S.**, Kamath, S., Santos, A. P. A., Bonidia, R. P., Espíndola-Hernández, P., Stadler, P. F., Oliveira, A., de Carvalho, A. C. P. L. F., & Rocha, U. N. Unveiling protein domains and families in 183,044 novel orthologs exploring multiple protein language models and over 16 million open reading frames recovered from thousands of terrestrial metagenomes. **Oral presentation**, Helmholtz AI Conference 2025, Messe Karlsruhe, Rheinstetten, Germany, 5 June 2025. Conference archive
- **de Almeida, B. L. S.**, Kamath, S., Santos, A. P. A., Bonidia, R. P., Espíndola-Hernández, P., Stadler, P. F., Wierckx, N., Borchert, E., Rother, D., Oliveira, A., de Carvalho, A. C. P. L. F., & Rocha, U. N. Different protein language models uncover 23.21% of the functional dark matter across 16 million open reading frames recovered from thousands of environmental metagenomes. **Oral presentation**, VAAM Annual Conference, Berlin, Germany, 22–25 March 2026 (SL-CM-01, Computational Microbiology). Abstract book.
- **de Almeida, B. L.**, Kamath, S., Santos, A. P. A., Bonidia, R. P., Espíndola-Hernández, P., Stadler, P. F., Oliveira, A., de Carvalho, A. C. P. L. F., & Rocha, U. N. Exploration of multiple protein language models and over 16 million open reading frames recovered from thousands of terrestrial metagenomes uncover protein domains and families in 13,486 previously unknown orthologs. In: 17th Symposium on Genetics and Bacterial Ecology (BAGECO 2025), 2025, Graz, Austria. **Flash talk**, 4 July 2025 (FT043). Abstract.

Conference Abstracts and Posters

- **de Almeida, B. L.**, Bonidia, R. P., Bole, M., Santos, A. P. A., Stadler, P. F., Rocha, U. N., & de Carvalho, A. C. P. L. F. From sequences to traits: An AI-driven AutoML framework for genome-resolved microbial ecology. In: 20th International Symposium on Microbial Ecology (ISME20), 2026, Auckland, New Zealand.
- Bonidia, R. P., Florentino, B. R., Barreto, M. C. A., **de Almeida, B. L.**, Constantino, J. L. R., Silva, D. T., Rocha, U. N., Goedert, G. T., Struchiner, C. J., & de Carvalho, A. C. P. L. F. AutoAI Hub: From Data to Society, Democratizing Machine Learning for Everyone. In: Human Genome Meeting 2026, 2026, Athens, Greece.
- Santos, A. P. A., **de Almeida, B. L.**, Bole, M., Kamath, S., & Rocha, U. N. A FAIR-by-Design and Standard-Compliant System for Automated Submission of Microbial Genomes to the European Nucleotide Archive. In: VAAM Annual Conference, 2026, Berlin, Germany.
- Bonidia, R. P., Santos, A. P. A., **de Almeida, B. L.**, Stadler, P. F., Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. P. L. F. BioAutoML: End-to-End Machine Learning Package for Life Sciences. In: 10th FEMS Congress of European Microbiologists, 2023, Hamburg, Germany.

Additional conference contributions and abstracts.

Teaching and Mentoring

2025 – 2026	**Instructor — UFZ Summer School**, Leipzig. Unix/Bash/Git, machine learning, and alignment-based and alignment-free annotation.
09.2025	**Instructor — NFDI4Microbiota workshop, ISAM**, Leipzig. Taught submission of research data and metadata to public repositories.
06.2025	**Co-organizer and instructor — BAGECO metagenomics workshop**, Graz. Taught experimental design, computer experiments, and end-to-end sequence analysis with BioAutoML.
05.2021 – 11.2022	**Teaching Assistant — University of São Paulo.** Eight hours/week across Object-Oriented Analysis and Design, Statistics, Fundamentals of Data Science, and Computational Visualization. Graded assignments, prepared exams, and delivered practical lectures.

Selected Professional Development

2025 **Summer School in Trends in Multi-omics Data Analysis for Microbial Ecology and Biotechnology**, UFZ, Leipzig.

2024 **15th Summer Course in Bioinformatics**, FMUSP, São Paulo.

2020 – 2021 Data science specializations: University of Michigan (*Applied Data Science with Python*) and Johns Hopkins University (*Data Science*), Coursera.

2020 *Nutrition and Health: Human Microbiome*, Wageningen University, edX.

Academic Service

Journal Peer Review

2026 – Present **Computational and Structural Biotechnology Journal.** One manuscript reviewed.

2026 – Present **Journal of Thermal Biology.** One manuscript reviewed.

2025 – Present **Briefings in Bioinformatics.** One manuscript reviewed across two rounds.

Undergraduate Thesis Examination

Member of six undergraduate thesis examination committees.

Community and Student Leadership

2018 – 2021 **Co-founder and coordinator — efestus**, University of São Paulo. Student organization addressing computing, data science, and technology ethics for social good. Funded participation in the TechShift Summit, Chapel Hill, USA.

2017 – 2019 **Software division member — Warthog Robotics**, University of São Paulo. Participated in robotics competitions and taught Git and object-oriented programming to new members.

Technical Skills

Methods Biological sequence representation; handcrafted descriptors and learned embeddings; automated machine learning; non-coding RNA classification; protein annotation; metagenomic sequence analysis.

Programming Python, R, Bash, Nix, SQL, Java, C, C++.

Modeling and data Scikit-learn, TensorFlow, pandas, Polars, NumPy, SciPy, NetworkX.

Sequence analysis Biopython, BLAST, Prodigal, OrthoFinder, Anvi'o, HMMER, EMBOSS, Infernal, DIAMOND, MMSeqs2.

Research tools Git, Linux, Jupyter, R Markdown, LaTeX, Typst, Matplotlib, Plotly, Streamlit.

Languages Portuguese (native), English (C1), German (A1.2).

Professional Memberships

International Society for Microbial Ecology (ISME), since October 2025; Federation of European Microbiological Societies (FEMS), since November 2025.